

Tutorial sheet 6B: Torsion of Circular Bar

1. A circular section bar of radius 20 mm and length 1 m is subjected to a torque of 100 N.m. Calculate the relative twist of the end, and, the max shear stress. ($G = 80 \text{ GPa}$).



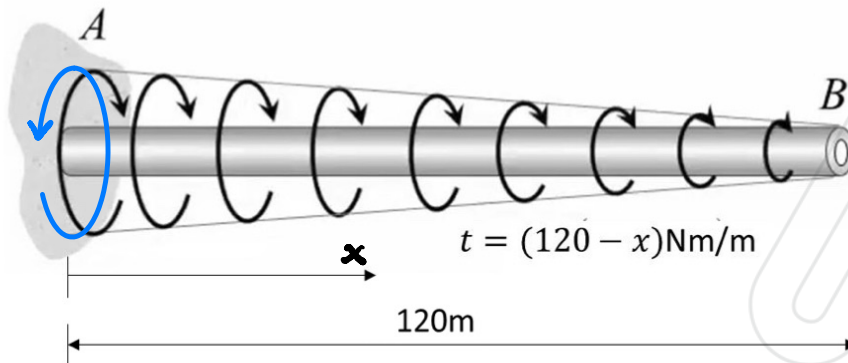
$$\therefore G \frac{d\theta}{dx} = \frac{T}{J} = \frac{\tau}{r}$$

$$\therefore \theta = \int \frac{T}{JG} dx = \frac{1}{JG} \int_0^1 100 dx = \frac{100}{JG}$$

$$\therefore J = \frac{\pi}{2} r^4 = 2.513 \times 10^{-7} \quad \therefore \theta = \frac{3.98 \times 10^8}{G}$$

$$\therefore \tau_{max} = \frac{Tr}{J} = \frac{2}{J} = 79.6 \text{ GPa}$$

2. For the hollow cylindrical shaft subjected to Torque as shown below determine the angle of twist at end B relative to the fixed end A. The inner radius $r_{in} = 1.5$ m, outer radius $r_{out} = 3$ m; $G = 3$ GPa.



$$T = 120x - x^2 \quad \sum T_x = 0 = T_A - \int_0^{120} (120 - x) dx \quad \therefore T_A = 1200$$

$$\therefore \phi_{AB} = \int_0^{120} \frac{\textcircled{1}}{JG} dx = \frac{1}{JG} \left[\overset{-\frac{1}{2}x^2 + 120x - 7200}{-\frac{1}{6}x^3 + 120x^2 - 7200x} \right]_0^{120} = \frac{\overset{-288000}{\textcircled{288000}}}{JG}$$

$$\therefore J = \frac{\pi}{2} (R_{out}^4 - R_{in}^4) = 119.28$$

$$\therefore \phi_{AB} \text{ at } \theta = \textcircled{3.81971 \times 10^{-6}} - 8.05 \times 10^{-7}$$